



Karnataka State Police
Government of Karnataka

Powered by **H2S**

0atathon 2026

Technology partner



Team Details

Team name: Team Zen

Team leader name: Hiran Vikraman S R

Team size: 2 (Hiran Vikraman S R · Arun G K)

Problem Statement: Challenge 1 — Intelligent Conversational AI for the KSP Crime Database. Enable officers to query the FIR/crime database in natural language (English & Kannada) and get accurate, evidence-grounded answers.



Brief about the solution

ANVESHAK (ಅನ್ವೇಷಕ) = “the investigator”) turns the KSP crime database into an **AI investigation partner** — not just a chatbot.

- Ask in Kannada or English, by voice or text — get an answer in seconds, **with the evidence** (SQL + rows + case IDs) attached. No SQL skills, no analyst queue.
- **Deterministic tools compute; the LLM only narrates** — every number traces to a tool result. Zero hallucinated statistics — essential for policing.
- It doesn't stop at answers — it works cases: links serial crime across districts, runs a 6-agent cell that assembles court-ready packs, traverses a crime knowledge graph, and sweeps overnight to raise leads before anyone asks.
- **Impact: minutes instead of days, catches siloed records miss, and proactive leads every morning. Runs 100% on Zoho Catalyst.**



Opportunities

How it is different, and how it solves the problem:

- Most “crime chatbots” stop at NL→SQL. ANVESHAK adds an investigation layer: cross-district serial linkage, multi-agent case work, and proactive patrol.
- **Trust by construction:** the LLM never invents numbers (deterministic tools + evidence drawer), and the linkage ground truth is public so results are independently verifiable.
- **Cross-district reach:** MO fingerprinting via multilingual embeddings finds serial offenders that siloed district records miss.
- **Built for real officers:** bilingual (Kannada + English) with browser voice, government-grade UI.

Where it makes a difference: a constable at a rural station gets a statewide answer in Kannada in seconds · an SP spots a serial ring crossing Mandya into Bengaluru before the next strike · an IO gets a first-draft, court-ready pack in minutes · a control room sees tomorrow’s hotspot tonight.

USP: “Ask a question, get a verified answer with evidence — and let the AI work the case end-to-end to a court-ready pack.”



List of features offered by the solution

- 1. Conversational floor** — bilingual EN + Kannada, voice + text, verified NL→SQL with evidence drawer on every answer.
- 2. Serial Crime Linkage Engine** — MO fingerprinting (multilingual embeddings + structured features), weighted cosine, HDBSCAN, 180-day / 120-km filter; discovers cross-district series cold.
- 3. AI Investigation Cell** — 6 agents stream their reasoning over SSE → court-ready Investigation Pack (PDF).
- 4. CrimeGraph** — knowledge graph over people/cases; ego-network, shortest path, Louvain communities for multi-hop questions.
- 5. Night Patrol** — spike / series-growth / repeat-offender detectors → ranked Lead Cards, autonomously.
- 6. Court-ready outputs** — Investigation Pack + conversation-to-PDF via SmartBrowz.
- 7. Governance** — RBAC scopes (SHO/SP/SCRB/analyst), audit log, and no protected attributes (religion/caste) in any model (ADR-9).



Process flow diagram or Use-case diagram

Officer (EN/KN, voice or text) → React SPA → FastAPI intent router:

- **NL→SQL path:** schema card + few-shots → sqlglot guardrails (SELECT-only, auto-LIMIT, self-repair) → DuckDB → answer + evidence drawer.
- **Tool path:** linkage / graph / forecast / hotspots / risk / similar → DuckDB → verified result → LLM narrates.
- **Investigation:** “Investigate SH-07” → 6-agent pipeline (SSE) → Investigation Pack → SmartBrowz PDF.
- **Autonomous:** Cron → Night Patrol detectors → Lead Cards → Signals / Mail digest.

Two data layers (ADR-1): Catalyst Data Store = system of record; an embedded DuckDB analytical mirror serves sub-second analytics.



Wireframes/Mock diagrams of the proposed solution (optional)

Six views in the live prototype (government-grade dark theme):

- **Chat** — bilingual answer + evidence drawer (SQL, rows, case IDs) + voice input.
- **Lead Feed** — ranked Night-Patrol Lead Cards.
- **Series** — discovered serial-crime hypotheses with confidence + linked case IDs.
- **CrimeGraph** — interactive people/case network with community highlighting.
- **Investigation Room** — six agent cards stream live, then the assembled pack.
- **Audit** — every action logged (user, role, action, detail).

Live: <https://anveshak-api-50044329134.development.catalystappsail.in/ui/>



Architecture diagram of the proposed solution

Officer (EN/KN, voice+text)

React SPA (served on Catalyst)

FastAPI backend on Catalyst AppSail → intent router

↳ NL→SQL (guardrails + self-repair) and typed tools

(linkage · graph · forecast · hotspots · risk · similar)

DuckDB analytical mirror ↔ Catalyst Data Store (system of record)

LLM adapter → Catalyst QuickML → GLM-4.7-Flash

6-agent Investigation Cell (SSE) → SmartBrowz → Pack PDF

Cron → Night Patrol detectors → Lead Cards

Deterministic tools, LLM narrates (ADR-2). No external inference API is ever called from the deployed app (ADR-4).



Technologies to be used in the solution

Backend: Python 3.12, FastAPI, DuckDB, sqlglot, sentence-transformers (paraphrase-multilingual-MiniLM-L12-v2), scikit-learn / HDBSCAN, NetworkX + Louvain, statsmodels SARIMA, sse-starlette.

Frontend: React 18, Vite, Tailwind CSS, ECharts (charts + network graph), Leaflet + OpenStreetMap, Web Speech API (kn-IN voice).

LLM: GLM-4.7-Flash served via Catalyst QuickML LLM Serving.

Packaging: Docker (custom AppSail runtime — bakes the scientific stack).

Quality: pytest suite (46 tests), bilingual eval harness (60 questions), public linkage ground truth.



List down Catalyst Services being used in the solution

Live now: AppSail (FastAPI, custom Docker runtime) · QuickML LLM Serving (GLM-4.7-Flash) · Web Client Hosting · Authentication · API Gateway.

In the architecture / provisioned: Data Store (system of record) · NoSQL (graph snapshots) · SmartBrowz (pack PDF) · Cron (Night Patrol) · Signals + Mail (lead digests) · Cache · Stratus (pack storage).

Deployment is exclusively on Zoho Catalyst, per the competition mandate.



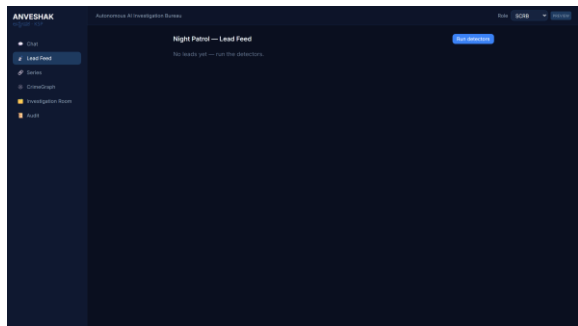
Estimated implementation cost (optional)

The prototype runs within the Catalyst development tier.

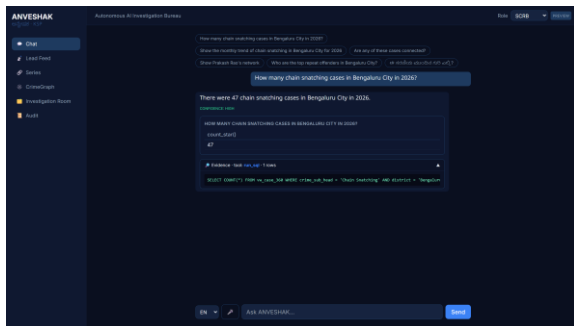
- AppSail: 1 GB memory, single instance (custom Docker runtime).
- QuickML: GLM-4.7-Flash is a Zoho-hosted shared model — no per-model deployment cost; billed per token at production scale.
- Web Client Hosting + Data Store + Cron: serverless, usage-based.
- Embeddings are precomputed at data-generation time, so runtime needs no GPU / heavy inference for search.



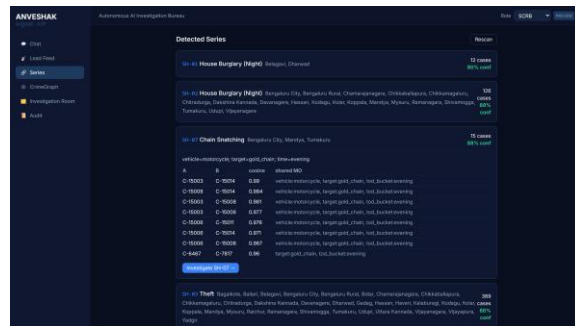
Snapshots of the prototype



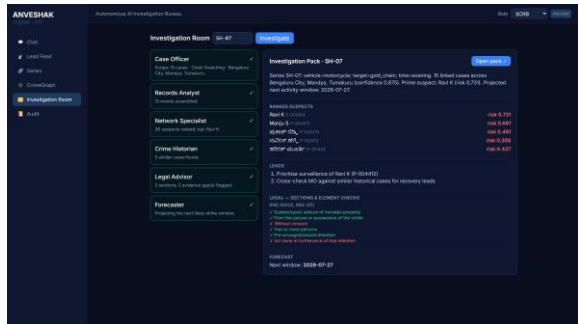
Night Patrol — proactive leads



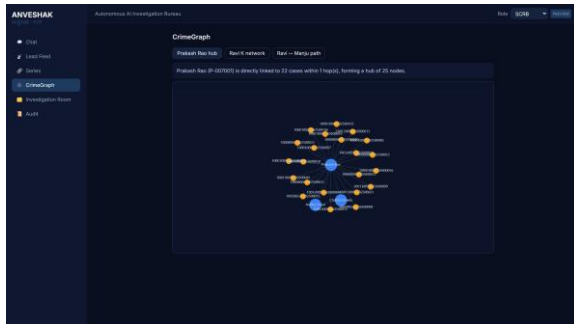
Chat — answer + evidence (SQL)



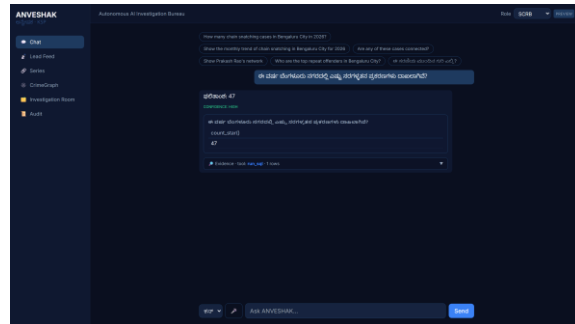
Serial linkage — SH-07



Investigation Pack — 6 agents



CrimeGraph — fraud hub



Bilingual — ಕನ್ನಡ + English



Prototype Performance report/Benchmarking

Dataset: 15,405 FIRs · 31 districts · 248 stations · 2023–2026 (synthetic, seed 42).

NL→SQL accuracy: 76.7% overall (EN 82.2%, KN 60.0%) on the local 7B dev model; gold-vs-gold 100%. Production GLM-4.7-Flash is expected higher.

Serial linkage (SH-07): precision 0.86 · recall 12/14; the disjoint SP-2 series correctly not merged.

Forecast (burglary): backtest MAE 1.83 vs seasonal-naive 1.75 — competitive and honestly reported.

Repeat-offender risk: Suresh B scores 0.83 (recency / frequency / gravity / centrality).

Night Patrol: Whitefield spike + repeat-offender cluster fire on the planted anomalies.

Robustness: thread-safe analytical layer — 60 concurrent requests, 0 errors; caches pre-warmed at startup.

Linkage ground truth is public (data_engine/planted/*.yaml) — numbers are independently verifiable.



Provide links to your:

GitHub (public): <https://github.com/arungeekay/anveshak>

Deployed link (Catalyst): <https://anveshak-api-50044329134.development.catalystappsail.in/ui/>

Demo video (3 min): (add your public YouTube / Drive link)

API base: <https://anveshak-api-50044329134.development.catalystappsail.in>



Additional Details/Future Development (if any)

Roadmap to production:

- RBAC server-side scope injection (designed, ADR-8) → full enforcement via Catalyst Auth.
- Live Data Store loader + FIR-intake write-back; NoSQL graph persistence.
- SmartBrowz pack PDF, Cron schedule, and Signals / Mail digests fully wired end-to-end.
- Real CCTNS identity resolution (name / parentage / DOB) replacing the synthetic person_key.
- Kannada NL→SQL tuning (hand-curated few-shots) to lift KN accuracy toward the EN level.
- Human-in-the-loop feedback on linkage / leads to continuously improve precision.



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THANK YOU